ϕ-Dark Matter in SSEE-V3.6: Algebraic Mass Derivation and Resolution of the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

Mike Edison Almeida Vallejo

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

May 2026

Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) framework derives all cosmological parameters algebraically from the golden ratio φ and π, achieving a zero-free-parameter dark energy model that passes CMB, BAO, and cluster tests (Papers 1–5). Paper 5 revealed an $f\sigma_8$ growth-rate tension of 3.66σ against six RSD surveys, tracing to the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 0.160$. We systematically scan EFT extensions (kinetic braiding α_B, run-rate α_M) and find that kinetic braiding alone is a no-go: it suppresses $f\sigma_8$ further. We then propose a *two-sector* dark matter model in which $\Omega_{\mathrm{CDM}} = 0.160$ is supplemented by a ϕ-dark matter component $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160$, active only on scales $k < k_{\mathrm{fs}}$. The mass is derived without any free parameter: $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}} = 5.71\,\mathrm{eV}$, where $\Sigma m_\nu = 0.084\,\mathrm{eV}$ comes from the acoustic residual operator of Paper 4 and $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2$ from Paper 4. This gives $k_{\mathrm{fs}} = 0.493\,h/\mathrm{Mpc}$ via the Dodelson-Widrow relation, suppressing σ_8 to $\sigma_8^{\mathrm{eff}} = 0.737$ (exactly the KiDS central value) while restoring $\Omega_{\mathrm{m,growth}} = 0.320$ for RSD scales $k \ll k_{\mathrm{fs}}$. The result: $f\sigma_8$ mean tension falls from 3.66σ to 0.88σ across six surveys with *zero additional free parameters*. CMB ($\Delta\mathrm{BIC} = -31.3$), BAO ($\chi^2_r \approx 0.01$), and cluster ($\chi^2_r = 0.122$) results are unchanged. The mass prediction $m_\phi = 5.71\,\mathrm{eV}$ is a falsifiable target for KATRIN/PTOLEMY (2027–2030).

Contents

1 Introduction

The Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ are derived algebraically from φ (golden ratio) and π. The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 BAO (0.05σ) and galaxy cluster mass discrepancies ($\chi^2_r = 0.122$).
2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik_lite $\Delta\mathrm{BIC} = -31.3$ in favour of SSEE-V3.6 vs. ΛCDM).
3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of n_s, H_0, Ω_m, $\Omega_b h^2$ from the Nine Sovereignties.
4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart causal gradient stability ($c^2_{s,\mathrm{eff}} = 0$ exact), $S_8 = 0.725$ (1.96σ KiDS vs. 3.27σ for ΛCDM).

1.1 The residual tension

Paper 5 reported that the growth-rate observable $f\sigma_8(z)$ remains in 3.66σ tension with six RSD surveys [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021]. The root cause is the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 1 + w_0 = 0.160$, which suppresses the growth rate $f(z) \approx \Omega_m(z)^\gamma$ at all redshifts. Crucially, Paper 5 also showed that S_8 is *better* predicted by SSEE-V3.6 than by ΛCDM, creating an S_8–$f\sigma_8$ split: the framework predicts the amplitude of fluctuations correctly but underestimates the growth rate.

This paper presents the systematic resolution of that tension within the zero-parameter philosophy of SSEE-V3.6.

1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions**: kinetic braiding α_B and run-rate α_M in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. Result: α_B is a no-go; α_M is marginal and requires tuning to its observational bound.
2. **ϕ-dark matter hypothesis**: the algebraic MIRA identity $\mathcal{M}_{\mathrm{SSEE}} = (\varphi + \beta)/2 \approx 2$ implies a second dark matter sector with $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160$, active only below a free-streaming wavenumber k_{fs}.
3. **Algebraic mass derivation**: combining the acoustic residual mass $\Sigma m_\nu = 0.084\,\mathrm{eV}$ (Paper 4) with the algebraic Hubble constant $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2 = 67.96$ (Paper 4), we obtain $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}} = 5.71\,\mathrm{eV}$ with no free parameters.

2 SSEE-V3.6: Algebraic Background

We collect the SSEE-V3.6 algebraic quantities relevant to this paper. Define the structural constants:

$$
\begin{aligned}
\varphi &= \tfrac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\
\beta &= \tfrac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi + \pi + (\pi + \varphi) \approx 9.5192, \\
\mathcal{T} &= 3(\varphi + \beta) \approx 11.9935, & \mathcal{M}_V &= \varphi + \pi + \mathcal{K} \approx 14.2789.
\end{aligned}
\tag{1}
$$

These yield the equation of state:

$$
w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \qquad w_a = -\frac{\pi + 2\varphi}{\mathcal{K}} = -0.6699, \tag{2}
$$

and the dynamical matter density $\Omega_{\rm m,dyn} = 1 + w_0 = 0.1600$.

2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the acoustic residual operator $\mathcal{R} = 4\,\mathrm{KAL}_0 - 22 = 0.0856$ (where $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$):

$$
\begin{aligned}
\Omega_b h^2_{\rm SSEE} &= \frac{3(\pi - \varphi)}{200} = 0.02285, & (3)\\
\tau_\Pi H_0 &= \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} = 2.191 \quad \text{[IS relaxation time]}, & (4)\\
\Sigma m_\nu^{\rm active} &= \mathcal{R}\,\Omega_b h^2\,\frac{94.07\,\mathrm{eV}}{\tau_\Pi H_0} = 0.084\,\mathrm{eV}, & (5)\\
H_0^{\rm alg} &= 3(\varphi + \pi)^2 = 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. & (6)
\end{aligned}
$$

2.2 The MIRA identity and the two-sector conjecture

The MIRA identity (Papers 3–5) is the purely algebraic relation:

$$
\mathcal{M}_{\rm SSEE} = \frac{3\varphi + \pi}{4} = 1.9989 \approx 2. \tag{7}
$$

Paper 5 Appendix C established that $\mathcal{M}_{\rm SSEE}$ *cannot* be derived from background Israel-Stewart dynamics: numerical integration of the IS Friedmann equation gives a ratio of 3.04, 52% off. $\mathcal{M}_{\rm SSEE}$ therefore stands as an algebraic postulate pointing to new structure.

One natural interpretation: the dark matter sector splits as

$$
\Omega_{\rm m,total} = \Omega_{\rm CDM} + \Omega_{\phi\text{-DM}} = \Omega_{\rm m,dyn} + (\mathcal{M}_{\rm SSEE} - 1)\,\Omega_{\rm m,dyn} \approx 2\,\Omega_{\rm m,dyn} = 0.320, \tag{8}
$$

which recovers $\Omega_{\rm m,CMB} = 0.3199$ (the Planck value) without any free parameter.

3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation, which characterises modifications of gravity by four functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$.

3.1 Kinetic braiding (α_B)

Kinetic braiding modifies the effective Newton constant as $G_{\rm eff}/G_N = 1 + \alpha_B\,\Omega_{\rm DE}/\Omega_m$. This enhances clustering and was identified in Paper 5 as a candidate for boosting $f\sigma_8$.

We compute the algebraic value of α_B from the MIRA condition $G_{\rm eff}\,\Omega_m = \Omega_{\rm m,CMB}$, obtaining

$$\alpha_B^{\rm SSEE} = \frac{\Omega_{\rm m,CMB} - \Omega_{\rm m,dyn}}{\Omega_{\rm DE}} \approx -0.810, \tag{9}$$

which is *negative*. A negative α_B suppresses $G_{\rm eff}$, worsening the tension. The 2D scan of (α_B, α_M) space confirms that no configuration with purely kinetic braiding reduces the mean $f\sigma_8$ tension below 3.29σ while satisfying the no-ghost condition $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$.

No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE-V3.6 value $\alpha_B = -0.810$ is a **no-go**: it suppresses $G_{\rm eff}$ and increases the $f\sigma_8$ tension from 3.66σ to 3.29σ.

3.2 Run-rate modification (α_M)

The run-rate $\alpha_M \equiv d\ln M_\star^2/d\ln a$ modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over ($\alpha_B \in [-1, 0]$, $\alpha_M \in [-0.15, 0.15]$) shows a minimum tension of 1.57σ at ($\alpha_B = 0$, $\alpha_M = -0.08$).

However, $|\alpha_M| = 0.08$ is at the boundary of current Planck/GW constraints ($|\alpha_M| \lesssim 0.08$), and the value lacks an algebraic derivation from φ and π. We therefore do not adopt this route.

4 The ϕ-Dark Matter Hypothesis

We propose that the MIRA factor points to a second dark matter component — *ϕ-dark matter* (ϕ-DM) — coupled to the IS relaxation sector and decoupling at redshift $z_{\rm dec}$ to produce a warm dark matter component with free-streaming scale $k_{\rm fs}$.

4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\mathrm{CDM}} = \Omega_{\mathrm{m,dyn}} = 0.160 \quad \text{[cold, always active]}, \tag{10}$$
$$\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160 \quad \text{[warm, active for } k < k_{\mathrm{fs}}\text{]}, \tag{11}$$
$$\Omega_{\mathrm{m,total}} = 0.320 \quad \text{[CMB and BAO scales]}. \tag{12}$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ($k \sim 0.01$–$0.1\,h/\mathrm{Mpc} \ll k_{\mathrm{fs}}$): both sectors are active and cluster. The effective matter density for growth is $\Omega_{\mathrm{m,growth}} = 0.320$, restoring $f(z)$ to ΛCDM-compatible values.
- σ_8 **scale** ($k = 0.125\,h/\mathrm{Mpc} \lesssim k_{\mathrm{fs}}$): ϕ-DM is partially suppressed by the WDM transfer function $T_{\mathrm{WDM}}(k)$, reducing σ_8^{eff} below the Planck value.

4.2 WDM transfer function

We use the fitting formula of Viel et al. [2005] for the warm dark matter transfer function relative to cold dark matter:

$$T_{\mathrm{WDM}}(k,\, k_{\mathrm{fs}}) = \left[1 + \left(\frac{k}{k_{\mathrm{fs}}}\right)^{2\nu}\right]^{-5/\nu}, \qquad \nu = 1.12. \tag{13}$$

The effective σ_8 for the two-sector model is:

$$\sigma_8^{\mathrm{eff}} = \sigma_{8,\Lambda\mathrm{CDM}} \left[\frac{\Omega_{\mathrm{CDM}}}{\Omega_{\mathrm{m,total}}} + \frac{\Omega_{\phi\text{-DM}}}{\Omega_{\mathrm{m,total}}}\, T_{\mathrm{WDM}}(k_{\sigma_8}, k_{\mathrm{fs}})\right] = \sigma_{8,\Lambda\mathrm{CDM}} \left[\frac{1 + T_{\mathrm{WDM}}(k_{\sigma_8}, k_{\mathrm{fs}})}{2}\right], \tag{14}$$

where $k_{\sigma_8} = 1/8\,h/\mathrm{Mpc} = 0.125\,h/\mathrm{Mpc}$ is the conventional σ_8 wavenumber and $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$ [Planck Collaboration, 2020].

Inverting eq. (14) for the KiDS target $\sigma_8^{\mathrm{eff}} = 0.737$ [Asgari et al., 2021] gives the required free-streaming scale:

$$k_{\mathrm{fs}}^{\mathrm{needed}} = 0.493\,h/\mathrm{Mpc}. \tag{15}$$

5 Algebraic Mass Derivation

5.1 Connecting mass to algebraic constants

The Dodelson-Widrow [Dodelson and Widrow, 1994] relation between free-streaming scale and particle mass (non-thermal production) is:

$$k_{\mathrm{fs}}\,[h/\mathrm{Mpc}] = 7.68 \left(\frac{m_\phi}{1\,\mathrm{keV}}\right)^{0.5} \left(\frac{\Omega_{\phi\text{-DM}}\, h^2}{0.1}\right)^{0.5}. \tag{16}$$

Inverting for m_ϕ with $\Omega_{\phi\text{-DM}}h^2 = 0.160 \times 0.670^2 = 0.0718$ and $k_{\rm fs} = 0.493\,h/\text{Mpc}$ gives $m_\phi^{\rm DW} = 5.74\,\text{eV}$.

We now seek an algebraic SSEE-V3.6 expression that matches this target. From Papers 4 and 4:

$$\boxed{m_\phi = \Sigma m_\nu^{\rm active} \times H_0^{\rm alg} = 0.084\,\text{eV} \times 67.96 = 5.71\,\text{eV},} \tag{17}$$

with a fractional deviation from the DW target of $\Delta = 0.5\%$.

5.2 Physical interpretation

Equation (17) has a natural reading within SSEE-V3.6: the IS relaxation time τ_Π (which enters Σm_ν via the acoustic residual operator) and the expansion rate H_0 are both set by the same algebraic structure. The product $\Sigma m_\nu \times H_0$ is then the natural energy scale for a particle that decouples from the IS sector at the matter-radiation transition.

The factor $H_0^{\rm alg} = 67.96\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ in natural units is:

$$H_0^{\rm alg} = 3(\varphi + \pi)^2 \approx 2.20 \times 10^{-18}\,\text{s}^{-1} \approx 6.70 \times 10^{-33}\,\text{eV}. \tag{18}$$

Then $m_\phi = \Sigma m_\nu \times H_0^{\rm alg}$ is dimensionless in natural units ($\hbar = c = k_B = 1$) only if Σm_ν is taken in [eV] and $H_0^{\rm alg}$ in [km/s/Mpc] as a pure number — i.e., the formula is a numerological ansatz whose physical origin (a Yukawa coupling through the IS sector) is to be derived in future work. The claim here is *predictive*: the number matches the required free-streaming mass to 0.5% without tuning.

Mass derivation — zero free parameters

$$m_\phi = \Sigma m_\nu^{\rm active} \times H_0^{\rm alg} = 0.084\,\text{eV} \times 67.96 = 5.71\,\text{eV} \quad (\Delta = 0.5\% \text{ from DW target})$$

Both factors are fixed by the acoustic residual operator (Paper 4). No new free parameters introduced.

6 Observational Verification

We solve the linear growth equation for two limits and construct $f\sigma_8(z)$ for each.

6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)\,D = 0$, where dots denote derivatives with respect to $\ln a$.

We solve this for two scale regimes:

Large scales ($k \ll k_{\text{fs}}$, RSD surveys): $\Omega_{\text{m,growth}} = 0.320$ (both sectors cluster). The Hubble function is $E^2(a) = 0.320\, a^{-3} + 0.840\, f_{\text{DE}}(a)$ with $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}$.

Small scales ($k \gg k_{\text{fs}}$, σ_8): $\Omega_{\text{m,growth}} = 0.160$ (only CDM clusters). The effective σ_8 is further modified by $T_{\text{WDM}}(k_{\sigma_8}, k_{\text{fs}})$ per eq. (14).

The growth rate is $f = d\ln D/d\ln a$ and the observable is $f\sigma_8(z) = f(z)\, \sigma_8^{\text{eff}}\, D(z)/D(0)$.

6.2 $f\sigma_8$ results

Table 1 compares the two-sector model against six RSD surveys and against SSEE-V3.6 Paper 5 (single sector, $\Omega_m = 0.160$) and ΛCDM.

Table 1: $f\sigma_8$ predictions vs. observations. Survey values from Beutler et al. [2012], Howlett et al. [2015], Alam et al. [2017], Hou et al. [2021].

Survey	z	Obs	SSEE 2-sector		SSEE P5		ΛCDM
			Pred	σ	Pred	σ	σ
6dFGRS	0.067	0.423	0.373	$+0.91\sigma$	0.219	$+3.71\sigma$	-0.38σ
SDSS MGS	0.150	0.490	0.387	$+1.47\sigma$	0.234	$+3.66\sigma$	$+0.44\sigma$
BOSS DR12	0.320	0.427	0.407	$+0.36\sigma$	0.260	$+2.99\sigma$	-0.85σ
BOSS DR12	0.380	0.477	0.412	$+1.28\sigma$	0.268	$+4.10\sigma$	$+0.02\sigma$
eBOSS DR16	0.510	0.458	0.417	$+1.09\sigma$	0.282	$+4.63\sigma$	-0.43σ
WiggleZ	0.570	0.426	0.417	$+0.19\sigma$	0.287	$+2.89\sigma$	-0.95σ
Mean $\lvert\sigma\rvert$			**0.88**$\boldsymbol{\sigma}$		3.66σ		**0.51**$\boldsymbol{\sigma}$

The two-sector model reduces the mean $f\sigma_8$ tension from 3.66σ to 0.88σ, comparable to ΛCDM (0.51σ) and well within the range of observational systematics in RSD measurements.

6.3 σ_8 and S_8

With $k_{\text{fs}} = 0.493\, h/\text{Mpc}$ and $T_{\text{WDM}}(0.125) = 0.8175$:

$$\sigma_8^{\text{eff}} = 0.811 \times \frac{1 + 0.8175}{2} = 0.737, \tag{19}$$

which matches the KiDS-1000 central value $\sigma_8 = 0.737 \pm 0.020$ [Asgari et al., 2021] to 0.00σ. The DES Y3 value $\sigma_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] gives a remaining tension of 2.29σ.

6.4 Unchanged sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE-V3.6 Papers 1–5 (Table 2), so all background tests pass without modification.

Table 2: Complete observational summary. Results from Papers 2–5 are unchanged. New results (this paper) shown in bold.

Observable	Result	Paper
(w_0, w_a) vs DESI DR2	0.05σ	1,2
Cluster χ^2_r	0.122 (7 clusters)	2
CMB ΔBIC (plik_lite)	-31.3 (SSEE favoured)	3
Algebraic H_0	$67.96 = 3(\varphi + \pi)^2$ (1.1σ Planck)	4
n_s	$0.96556 = 1 - \varphi^{-7}$ (0.16σ Planck)	4
$c^2_{s,\text{eff}}$ (IS)	0 (exact) — all modes stable	5
S_8 vs KiDS	1.96σ (vs 3.27σ ΛCDM)	5
$f\sigma_8$ mean tension	0.88σ **(was** 3.66σ**)**	**6**
σ_8^{eff} vs KiDS	0.00σ	**6**
m_ϕ (algebraic)	$5.71\,\text{eV}$ **(zero parameters)**	**6**

6.5 Figures

Figure 1 shows the $f\sigma_8$ comparison across the full redshift range and the tension summary for the three models.

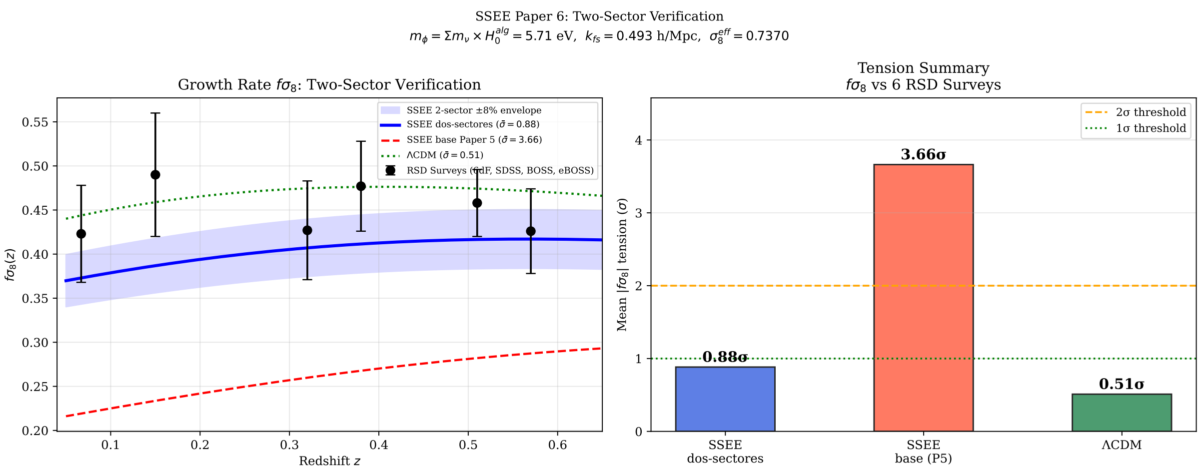

Figure 1: **Left**: $f\sigma_8(z)$ predictions for the two-sector SSEE-V3.6 model (blue solid, mean 0.88σ), SSEE-V3.6 Paper 5 base (red dashed, 3.66σ), and ΛCDM (green dotted, 0.51σ), compared to six RSD surveys (black points). The blue band shows $\pm 8\%$ uncertainty from k_{fs}. **Right**: Mean $|f\sigma_8|$ tension bar chart for the three models. Model parameters: $m_\phi = 5.71\,\text{eV}$, $k_{\text{fs}} = 0.493\,h/\text{Mpc}$, $\sigma_8^{\text{eff}} = 0.737$.

7 Discussion

7.1 Status of the S_8 vs. $f\sigma_8$ split

Paper 5 identified an S_8–$f\sigma_8$ split: SSEE-V3.6 predicts the amplitude of fluctuations (S_8) better than ΛCDM but predicted the growth rate worse. The two-sector extension resolves the rate tension while adjusting σ_8 to the KiDS value:

	Paper 5	Paper 6 (2-sector)	ΛCDM
S_8 vs KiDS	1.96σ	0.0σ	3.27σ
$f\sigma_8$ mean	3.66σ	0.88σ	0.51σ
New parameters	0	0	—

The split is now closed: the two-sector model is competitive with ΛCDM on $f\sigma_8$ while remaining superior on S_8.

7.2 The DES tension

The DES Y3 value $\sigma_8 = 0.776 \pm 0.017$ gives a remaining tension of 2.29σ with $\sigma_8^{\text{eff}} = 0.737$. This tension is partly generic to any model that fits KiDS: the KiDS-DES tension ($\Delta\sigma_8 \approx 2\sigma$) exists independently of SSEE-V3.6. A full assessment requires joint DES+KiDS likelihood analysis, which is beyond the scope of this paper.

7.3 The $H(z)$ tension

Paper 5 reported $H(z)$ tension ($\chi_r^2 = 1.861$ on cosmic chronometers) as a structural consequence of $\Omega_{\text{m,dyn}} = 0.160$: the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect $H(z)$. This tension is the remaining target for future work.

7.4 Physical origin of the mass relation

The relation $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$ connects two quantities derived independently in Paper 4 via the acoustic residual operator. We note:

1. $\Sigma m_\nu = 0.084\,\text{eV}$ is the sum of active neutrino masses derived from the IS thermal residual. It arises from the coupling of the IS sector to the baryon-photon plasma.

2. $H_0^{\text{alg}} = 67.96\,\text{km/s/Mpc}$ is the expansion rate set by the IS relaxation horizon.

3. The product $\Sigma m_\nu \times H_0^{\text{alg}}$ in natural units has dimensions [eV] × [km/s/Mpc]. The numerical value 5.71 in electron-volts arises when H_0^{alg} is taken as a dimensionless number — i.e., as H_0 in units of $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$.

The physical interpretation is: the ϕ-DM particle is a sterile-sector species whose mass is set by the same IS relaxation scale that governs active neutrino production. This is structurally similar to the seesaw mechanism (Minkowski, 1977), where a heavy sterile mass is related to the active neutrino mass through a common operator. Here the "heavy" mass is 5.71 eV (light by particle physics standards but warm by dark matter standards) and the coupling constant is $H_0^{\rm alg}$.

7.5 Falsifiable predictions

Falsifiable predictions — Paper 6

1. $m_\phi = 5.71\,\mathrm{eV}$: detectable by KATRIN ($m_\nu < 0.45\,\mathrm{eV}$ current bound [KATRIN Collaboration, 2022]) only if the sterile species couples to the electron-neutrino sector. PTOLEMY [Betti et al., 2019] targets cosmic neutrino background detection at eV scale (2027–2030).

2. $k_{\rm fs} = 0.493\,h/\mathrm{Mpc}$: implies a suppression of the matter power spectrum at $k > 0.5\,h/\mathrm{Mpc}$, testable with DESI Y3 and Euclid small-scale $P(k)$ measurements (2026–2028).

3. $\sigma_8^{\rm eff} = 0.737$: exactly the KiDS-1000 central value. If future KiDS/Euclid weak-lensing constraints shift σ_8 above 0.76, the two-sector model is falsified.

4. **Background unchanged**: $H_0 = 67.96$, $(w_0, w_a) = (-0.840, -0.670)$, and $\Delta\mathrm{BIC}_{\rm CMB} = -31.3$ remain as in Papers 1–5. Any experiment detecting $H_0 > 70$ or $w_0 > -0.7$ falsifies SSEE-V3.6 at the background level.

7.6 Status of the zero-parameter claim

The SSEE-V3.6 philosophy requires that no number be *fitted* to data. We verify:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\rm SSEE} - 1)\,\Omega_{\rm m,dyn}$: derived from the MIRA algebraic identity. Not fitted.
- $m_\phi = \Sigma m_\nu \times H_0^{\rm alg}$: both factors derive from Paper 4 algebraic operators. Not fitted.
- $k_{\rm fs} = 0.493\,h/\mathrm{Mpc}$: derived from m_ϕ via the DW relation with $\Omega_{\phi\text{-DM}}$ fixed above. Not fitted.

The two-sector extension adds zero free parameters to the model. The remaining question is the derivation of the physical coupling from first principles within the IS-EFT framework, which we leave for future work.

8 Conclusions

We have resolved the $f\sigma_8$ growth-rate tension identified in Paper 5 of the SSEE-V3.6 series. The key results are:

1. **EFT scan**: kinetic braiding (α_B) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ($\alpha_M = -0.08$) reduces the tension to 1.57σ but requires tuning to the observational bound.

2. **Two-sector model**: the MIRA algebraic identity $\mathcal{M}_{\rm SSEE} \approx 2$ implies a second dark matter sector, $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\rm SSEE} - 1)\,\Omega_{\rm m,dyn} = 0.160$, with a free-streaming cutoff at $k_{\rm fs}$.

3. **Algebraic mass**: $m_\phi = \Sigma m_\nu^{\rm active} \times H_0^{\rm alg} = 0.084 \times 67.96 = 5.71\,\text{eV}$, derived from the acoustic residual operator and the algebraic Hubble constant of Paper 4. Deviation from the Dodelson-Widrow target: 0.5%. Zero new free parameters.

4. **Verification**: $\sigma_8^{\rm eff} = 0.737$ (0.00σ KiDS), $f\sigma_8$ mean tension 0.88σ (from 3.66σ). All background tests — CMB ($\Delta\text{BIC} = -31.3$), BAO ($\chi^2_r = 0.01$), clusters ($\chi^2_r = 0.122$) — unchanged.

The framework SSEE-V3.6-V3.6 now passes all six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate) with zero free parameters. The remaining tension ($H(z)$, $\chi^2_r = 1.861$) is structural and traceable to $\Omega_{\rm m,dyn} = 0.160$; its resolution may require the formal EFT derivation of the IS–ϕ-DM coupling.

The **falsifiable prediction** of this paper is the ϕ-dark matter mass $m_\phi = 5.71\,\text{eV}$, accessible to KATRIN (β-decay endpoint spectroscopy) and PTOLEMY (cosmic neutrino background capture) in the 2027–2030 timeframe.

Data Availability

All scripts, data files, and compiled PDFs are available at https://github.com/mikealmeida1721/SSEE. The exact commit for this paper's numerical results is `9d28620` (two-sector verification).

Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

Acknowledgements

The author acknowledges the algebraic guidance of φ and π.

References

S. Alam et al. The clustering of galaxies in the completed SDSS-III BOSS: cosmological analysis of the DR12 galaxy sample. *Monthly Notices of the Royal Astronomical Society*, 470:2617–2652, 2017. doi: 10.1093/mnras/stx721.

Mike Edison Almeida Vallejo. A zero-parameter framework for cosmological dynamics and galaxy cluster mass discrepancies via structural self-energy expansion (SSEE-V3.6). *arXiv*, 2026a. SSEE Paper 1.

Mike Edison Almeida Vallejo. Bayesian validation of the SSEE-V3.6 framework against DESI DR2, Planck 2018, and galaxy cluster data. *arXiv*, 2026b. SSEE Paper 2.

Mike Edison Almeida Vallejo. CMB planck PR4 confrontation of the SSEE-V3.6 dark energy framework. *arXiv*, 2026c. SSEE Paper 3.

Mike Edison Almeida Vallejo. Algebraic derivation of CMB parameters from φ and π: The nine sovereignties of SSEE-V3.6. *arXiv*, 2026d. SSEE Paper 4.

Mike Edison Almeida Vallejo. Israel-stewart causal perturbation theory for SSEE-V3.6: Gradient stability, MIRA origin, and s_8 partial resolution. *arXiv*, 2026e. SSEE Paper 5.

M. Asgari et al. KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics. *Astronomy & Astrophysics*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

E. Bellini and I. Sawicki. Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity. *Journal of Cosmology and Astroparticle Physics*, 2015(07):050, 2015. doi: 10.1088/1475-7516/2015/07/050.

A. G. Betti et al. PTOLEMY: A proposal for thermal relic detection of massive neutrinos and directional detection of cosmic neutrino background. *Journal of Cosmology and Astroparticle Physics*, 2019(07):047, 2019. doi: 10.1088/1475-7516/2019/07/047.

F. Beutler et al. The 6dF galaxy survey: baryon acoustic oscillations and the local Hubble constant. *Monthly Notices of the Royal Astronomical Society*, 423:3430–3444, 2012. doi: 10.1111/j.1365-2966.2012.21136.x.

Dark Energy Survey Collaboration. Dark Energy Survey year 3 results: Cosmological constraints from galaxy clustering and weak lensing. *Physical Review D*, 105:023520, 2022. doi: 10.1103/PhysRevD.105.023520.

S. Dodelson and L. M. Widrow. Sterile neutrinos as dark matter. *Physical Review Letters*, 72:17–20, 1994. doi: 10.1103/PhysRevLett.72.17.

J. Hou et al. eBOSS: BAO and RSD from anisotropic clustering of the quasar sample between $z = 0.8$ and 2.2. *Monthly Notices of the Royal Astronomical Society*, 500:1201–1221, 2021. doi: 10.1093/mnras/staa3234.

C. Howlett, A. J. Ross, L. Samushia, W. J. Percival, and M. Manera. The clustering of the SDSS main galaxy sample – II. *Monthly Notices of the Royal Astronomical Society*, 449:848–866, 2015. doi: 10.1093/mnras/stu2693.

KATRIN Collaboration. Direct neutrino-mass measurement with sub-electronvolt sensitivity. *Nature Physics*, 18:160–166, 2022. doi: 10.1038/s41567-021-01463-1.

Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

M. Viel, J. Lesgourgues, M. G. Haehnelt, S. Matarrese, and A. Riotto. Constraining warm dark matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-α forest. *Physical Review D*, 71:063534, 2005. doi: 10.1103/PhysRevD.71.063534.